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Examiners' Report Summer 2026

Pearson Edexcel GCE

In Further Mathematics (9FM0)

Paper 3C Further Mechanics 1

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General comments

Learners were able to access all seven questions on this paper. Learners were able to recall and use standard formulae and were familiar with the context given in all questions.

If there is a printed answer to show, then learners should show sufficient detail in their working to warrant being awarded all the marks available and that they ensure that their final answer is exactly the same as the printed answer.

In all cases, as stated on the front of the question paper, learners should show sufficient working to make their methods clear to the examiner and correct answers without working may not score all, or indeed, any of the marks available.

In questions, the numerical value of g which should be used is $9.8\ \mathrm{ms}^{-2}$. Final answers should then be given to 2 (or 3) significant figures – more accurate answers will be penalised, including fractions but exact multiples of g are usually accepted.

Comments on individual questions

Question 1

The opening question to the paper examined an elastic string in equilibrium. The first 3 marks were awarded for resolving forces to find two relevant force equations. Although slips in sin/cos use was condoned, the responses who resolved lengths instead of forces were unable to access these marks. The following 2 marks were independent and for the use of Hooke's Law. It was rare to see method errors here but some used *L* for the natural length of the string instead of 2. The final method mark was dependent on a complete method leading to at least one of the correct values. Those responses who used an incorrect method, such as resolving lengths instead of forces, did not access these marks.
It is important for learners to notice when they replace g with 9.8 ms^{-2} because subsequent answers must be given to 2 (or 3) significant figures to earn all the available marks.

Question 2

Part (a)
The opening to this question on work-energy asked learners to show a given answer. The context for the question was a particle sliding down a rough inclined plane. Learners were expected to present their initial energy terms in a work-energy equation, followed by at least one line of working, to reach the given answer. This was answered well, with some learners presenting answers exactly as stated in the question. Throughout this question, the formulae for work-done, kinetic energy and gravitational potential energy were familiar.

Part (b)
The preferred method to find the value of μ was to divide the given value for work-done, by $2R$. Although a longer process, some learners chose to use the *suvat* equations to find the acceleration and then N2L to find μ. Both approaches appeared to be equally successful. However, the responses that used $R = 2g$, instead of the weight component, were unable to make progress. It is important for learners to notice when they replace g with 9.8 ms^{-2} because subsequent answers must be given to 2 (or 3) significant figures to earn all the available marks.

Part (c)
In this part of the question, the particle moves along a horizontal surface. After finding the distance travelled along the horizontal surface, it is necessary to add 2 to find the total distance travelled. Learners who continued to use the same friction value, were unable to form a valid equation and could not access any further marks.

Question 3

In each part of this question, learners were expected to replace g with $9.8\ \mathrm{ms}^{-2}$ and give numerical answers that were correctly rounded to 2 (or 3) significant figures. It is a good exam technique to double check that values have been copied down from the question correctly and used consistently throughout a question.

The most common incorrect values to appear in parts of the working were: 1000kg for the mass of the car and 1500W for the power of the engine.

Part (a)

The opening part tested understanding of the Power equation and N2L for a car and trailer moving up a slope. Learners confidently applied both equations to the given context.The successful responses formed a whole-system equation for N2L, instantly forming an equation in terms of the driving force. Those reponses who formed two equations of motion earned the first method mark when T had been eliminated and an equation had been formed in D alone. Occasional slips were seen when using an extended process.

Part (b)

The most successful method to find the tension in the tow bar, was to form an equation of motion for the trailer. Those responses who used the equation of motion for the car, instead of the trailer, required a correct driving force, found in part (a), to obtain the correct tension.

Part (c)

To complete the question, learners were required to form a work-energy equation for the car and trailer whilst the brakes of the car were applied. To earn method marks, energy equations must contain all the relevant terms with no additional terms. A common reason for not gaining the method marks in this part of the question was for having additional terms in the energy equation due to double-counting (including both the work-done against weight and GPE).

Question 4

Part (a)

Most learners produced a neat diagram with clearly labelled components before forming equations for CLM and NEL along the line of centres. It was common for learners to use column vectors in their working such as $\binom{5}{4}$, but most gave final answers using correct **i**-**j** notation as required.

Solutions generally demonstrated fluency in this topic with the most common errrors being:

- Incorrect components labelled on a diagram, despite a correct column vector being stated i.e. 4 labelled in the **i** direction and 5 labelled in the **j** direction.
- Incorrect use of column vectors for NEL.

  For example, $e = \frac{\binom{x}{y}-\binom{v}{w}}{\binom{2}{-2}-\binom{5}{4}}$ is an incorrect method.

Part (b)

Most learners found the initial kinetic energy and multiplied by 0.95 to account for a 5% KE loss. After forming a suitable equation in e, learners solved to find $e = 0.5$. It was possible to calculate the value 0.5 from incorrect working and in these cases, a maximum of 2 marks could be gained. Some answers followed from processing slips or sign errors. If 0.95 was incorrectly used by only considering kinetic energy along the line of centres, then marks could not be awarded in this part.

Question 5

Part (a)

There were 2 marks available in this part and most learners used $\frac{1}{2}$ correctly with Pythagoras to find an expression for $a^2 + b^2$. A common reason for not gaining marks was using $\frac{1}{2}$ on the wrong side of the equation or the incorrect assumption that $a = 1$ and $b = \frac{\lambda}{2}$. The latter produced the required result but did not earn the marks as it followed from an incorrect method.

Part (b)

The most efficient approach to this part used the scalar product to produce an equation in a, b and λ in just 2 lines of working. The two most common alternatives used perpendicular gradients or parallel vectors. These solutions were generally less successful than those using the scalar product.

Part (c)

This part scored well. To find the possible values of λ, learners were expected to find an expression for impulse first, and then use Pythagoras to form an equation in a, b and λ Although most learners used their earlier working to eliminate a and b, some responses started afresh and produced a fully correct solution. Those who incorrectly assumed $a = 1$ and $b = \frac{\lambda}{2}$, found the relevant λ values but could not earn the final mark due to their incorrect approach.

Question 6

Part (a)

The opening part to this question on successive, direct collisions, was scoring well, with learners stating equations for CLM and NEL before solving them simultaneously to obtain the given answer. Learners understood the importance of writing their answer in the given form. Occasionally learners wrote down, and used, the reciprocal fraction for NEL which is an incorrect method.

Part (b)

The most common reason for not gaining marks, when using the impulse-momentum equation, is an algebraic sign slip. This results in a maximum score of 1 mark out of the 3 available. The second most common reason for not gaining marks is using an incorrect method for mass. For those responses who used the impulse-momentum equation correctly, the final mark was awarded for a correctly factorised expression. It is recommended to look for common factors early on, such as $(e + 1)$, rather than expanding first and then attempting to factorise.

Part (c)

To find the range of possible values of e, it was necessary to find the speed of *P* after the collision. Taking account of the information that *P* reversed direction, produced $|v_P| = \frac{u}{5}(4e - 1)$ and the inequality $e > \frac{1}{4}$. For a second collision to occur between *P* and Q , the inequality $|v_Q| \geq |v_P|$ led to $e < \frac{1}{3}$, resulting in $\frac{1}{4} e < \frac{1}{3}$. Some found one of the values and assumed the other end of the inequality would be 0 or 1. It is important to check the information given in the question carefully to ensure all conditions are satisfied.

Question 7

There were many alternative approaches to this question on oblique impact and some learners answered part (b) within their solution to part (a). Those who took the given information and continued to work with vectors were able to make progress, however, there were responses where tried to convert the information into speeds and angles and then back into vectors. Some learners attempted to use a matrix approach and sometimes this was unsuccessful.

The most successful learners applied $\mathbf{u}.\hat{\mathbf{d}} = \mathbf{v}.\hat{\mathbf{d}}$ and $e = -\frac{\mathbf{v}.\hat{\mathbf{I}}}{\mathbf{u}.\hat{\mathbf{I}}}$ where $\mathbf{I}$ is the impulse direction and $\mathbf{d}$ is the direction perpendicular to impulse. Any multiple of these equations are acceptable. These formulae quickly produced two relevant equations in p , q and e which could be combined with the information about the loss in kinetic energy to answer both parts in a neat, efficient solution.

Part (a)

Learners were expected to answer this part by producing two simultaneous equations in p and q . First, recognising that the velocity component parallel to the wall was unchanged, produced $8 = -p + 2q$. Second, correct use of 0.25 to take account of the loss in KE, produced $p^2 + q^2 = 13$. Solving these led to two possible pairs of values for p and q. This question was a 'show that' question which asked for the answer to be carefully justified. Learners were expected to justify why the given values, -2 and 3, were the only possible solution. In most cases, this required showing that -2 and 3 satisfied the physical context **and** that the values -1.2 and 3.4 did not. Some learners found the corresponding e value for both pairs, providing a good justification and answering part (b) at the same time.

Apart from processing errors and insufficient justification, the most common reason for not gaining marks was use of 0.75 instead of 0.25 for the loss in KE.

Part (b)

Learners who were unable to find the values for p and q in part (a), were still able to gain all 3 marks in part (b) due to the given answers.

Use of $e = \pm\frac{\mathbf{v},\hat{\mathbf{I}}}{\mathbf{u}.\hat{\mathbf{I}}}$ was commonly seen, with the modulus being taken by those responses which did not include the minus sign in their formula.

The most common incorrect approach was use of $e = \frac{|\mathbf{v}|}{|\mathbf{u}|}$ which was an incorrect method and therefore gained no marks.